

§11.6 - Absolute Convergence and the Ratio and Root Tests

Definition #1 (Absolute Convergence)

Let (a_n) be a sequence.

If

$$\sum |a_n|$$

converges then the series $\sum a_n$ **converges absolutely**.

Definition #2 (Conditional Convergence)

Let (a_n) be a sequence.

If

$$\sum |a_n| = \infty$$

diverges but

$$\sum a_n$$

exists then the series $\sum a_n$ **converges conditionally**.

Example #1

The series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{\sqrt{k}}$$

converges conditionally while

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2 + \sqrt{k}}$$

converges absolutely.

Theorem #1

Suppose $\sum a_n$ converges absolutely.

Then $\sum a_n$ converges.

Example #2

We showed in §11.5:

$$\ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

$$\frac{1}{2} \ln(2) = \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \dots$$

Theorem #2 (Riemann's Rearrangement Theorem)

Let $x \in \mathbf{R}$.

Suppose

$$S = \sum_{k=1}^{\infty} a_k$$

converges conditionally.

Then we can rearrange the terms in the series so that the sum of the reordered terms converges to x .

Warning

The reordering of the terms in an absolutely convergent series does not change the value of the sum.

Definition (Geometric Series)

Let $x \in \mathbf{R}$.

The **geometric series**

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x}$$

exists only for $|x| < 1$.

Theorem #3 (Root Test)

Let (a_n) be a sequence and define

$$L = \lim_{n \rightarrow \infty} |a_n|^{1/n}.$$

Then

1. $L < 1$ implies $\sum a_n$ converges absolutely
2. $L > 1$ implies $\sum a_n$ diverges
3. $L = 1$ is inconclusive so use another test.

Example #3

Determine the convergence of

$$\sum \left(\frac{4n^2 - 3}{7n^2 + 6} \right)^n .$$

Example #4

Determine the convergence of

$$\sum \frac{2^n}{n^{10}}.$$

Theorem #4 (Useful Limits)

1. $n^{1/n} \rightarrow 1$

2. $|a|^{1/n} \rightarrow 1$ for $a \neq 0$

3. $\left(1 + \frac{x}{n}\right)^n \rightarrow e^x$

Definition #3 (Factorial)

Let $n \in \mathbf{N}$.

Then

$$\begin{aligned} n! &= n \times (n - 1) \times (n - 2) \times \cdots \times 3 \times 2 \times 1 \\ &= n(n - 1)! \end{aligned}$$

defines the n **th factorial**.

Theorem #5 (Ratio Test)

Let (a_n) be sequence of non-zero numbers and define

$$L = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}.$$

Then

1. $L < 1$ implies $\sum a_n$ converges absolutely
2. $L > 1$ implies $\sum a_n$ diverges
3. $L = 1$ is inconclusive so use another test.

Warning

Sums of powers of n will often not work with the root or ratio test.

Example #5

Determine the convergence of

$$\sum \frac{n^2}{n!}.$$

Example #6

Determine the convergence of

$$\sum \frac{n^n}{n!}.$$

Remark

- Root Test: good for powers of n
- Ratio Test: good for $n!$ or recursive terms
- Root Test is stronger than the Ratio Test, it works for more series. (Take Math 104)